Recitaion 04/05

While/For loop

W stands for while: these exercises have to be solved with while loops
F stands for for: these exercises have to be solved with for loops

Exercise 1 | Predict the output (W)

```
N = input('Enter a string: ')
i, j = 0, 0
while i < len(N):
    j = i + 1
    while j < len(N) and N[j] == N[i]:
        j = j + 1
    print( N[i], end='' )
    i = j
```

Input	Expected output
abcdeabcdeabcde	
aabbccddeeff	
abcaabbccaaabbbcccddd	

Exercise 2 | Calories Burned (W/F)

Running on a particular treadmill you burn `4.2` calories per minute.
Write a program that uses a loop to display the number of calories burned after `10, 15, 20, 25, and 30` minutes.

Exercise 3 | Fibonacci (W/F)

Write a program that asks the user to input a strictly positive integer value `n,` and displays the Fibonacci sequence up till the nth value.

The Fibonacci numbers may be defined by the following recurrence relation:

$$n = 0, F_0 = 0$$

$$n = 1, F_1 = 1$$

$$n > 1, F_n = F_{n-1} + F_{n-2}$$

Exercise 4 | Guessing Competition (W/F)

Write a Python program that plays a number guessing game.
The computer randomly chooses an integer between `1` and `100`. The player has at most `7` guesses to find the secret number.
After each guess:

- If the guess is too high, display "`Too high!`"
- If the guess is too low, display "`Too low!`"
- If the guess is correct, display "`Correct!`" and the number of guesses used.

After an incorrect guess, also display the current possible range of the secret number.
If the player does not guess correctly within `7` attempts, display "`Game over`!" and reveal the secret number.

Exercise 5 | Repetition of a substring (W)

Write a program that outputs the number of times a string appears into another.
Example output:

```
>>>Write a string: abcdeabcdeabcde
>>>Write another string: abc
      String ‘abc’ appears 3 times.

>>>Write a string: aaabcaa
>>>Write another string: aa
      String ‘aa’ appears 2 times.
```

Exercise 6 | Greatest Common Divisor (W)

The greatest common divisor of two positive integers, `n` and `m`, is the largest number, `d`, which divides evenly into both `n` and `m`. There are several algorithms that can be used to solve this problem, including:

- Initialize `d` to the smaller of `m` and `n`.
  While `d` does not evenly divide `m` or `d` does not evenly divide `n` do
- Decrease the value of `d` by 1
- Report `d` as the greatest common divisor of `n` and `m`

Write a program that reads two positive integers from the user and uses this algorithm to determine and report their greatest common divisor.

Exercise 7 | Predict the output (F)

Program 1	Output 1

`sum = 0`
`for i in range(20,2,-2):`
`  sum += i`
`print(sum)`	

Program 2	Output 2

`for i in range(0, 3):`
`    print(i+1, end=" ")`
`    for j in range(3, i, -1):`
`        print(j+1, end=" ")`
`    print()`	

Program 3	Output 3

`sum = 0`
`for i in range(1,15):`
`    if '4' in str(i):`
`        continue`
`    if '9' in str(i):`
`        break`
`    sum += i`
`    print(i,sum)`	

Program 4	Output 4

`result = 1`
`for c in "123":`
`    for d in "321":`
`        result *= int(c)`
`        result //= int(d)`
`print(result)`	

Program 5	Output 5

`s, t = "12345", 0`
`for i in range(1,6):`
`    t += int(s[len(s)-i])`
`    print(t)`	

Exercise 8 | Patterns (F)

1) Write a program that uses nested loops to draw this pattern:

```
>>>Enter the numbers of rows: 5
```

```
*****
****
***
**
*
```

2) Write a program that uses nested loops to draw this pattern:

```
>>>Enter the numbers of rows: 6
```

```
##
# #
#  #
#   #
#    #
#     #
```

3) Write a program that uses nested loops to draw this pattern:

```
>> Enter the number of rows: 8
```

```
        A
       B C
      D E F
     G H I J
    K L M N O
   P Q R S T U
  V W X Y Z A B
 C D E F G H I J
```

4) Write a program that draws N triangles of size M (both values are input by the user).

```
>>> Enter the number of triangles: 3
```

```
>>> Enter the size of one triangle: 3
```

```
*
**
***
*
**
***
*
**
***
```

Exercise 9 | Dice statics (F)

Write a program that rolls a six-sided die 1000 times.
Keep track of the numbers of ones, twos, . . . , and sixes that got rolled and display them.

Exercise 10 | Theme Park Ride Admission (W/F)

Ten people are waiting to ride a roller coaster. The roller coaster has only 5 available seats. For each person, ask for their height in centimeters.
Rules:

- A person must be at least `120 cm` tall to ride.
- If a person is too short, print `"Sorry, you are not tall enough."` and skip to the next person.
- If the person is tall enough, allow them to board.
- Once 5 people have boarded, stop checking the remaining people.

At the end, display the number of passengers. Example:

```
>>Person 1 - Enter height: 135
Welcome aboard!
>>>Person 2 - Enter height: 110
Sorry, you are not tall enough.
...
The ride is full!
5 passengers are on board.
```

Exercise 11 | Password Generator (F)

Write a program that repeatedly generates passwords for the user. Upon every iteration, the program starts by asking the user to input a password length. If the user inputs 0, the program terminates. If the value X input by the user is greater than 0, the program creates a password comprising X numerical digits chosen at random between 0 and 9.
Example output:

```
>>>Password size? > 4
9605
>>>Password size? > 9
654395954
>>>Password size? > 1
4
>>>Password size? > 0
```